Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0.333... (the 3 repeats forever) a rational number? Student says: No

Decide if the decimal repeats or ends.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A repeating decimal can be written as a fraction. 0.333... = 1/3, so it is rational.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $\sqrt{2}$ a rational number? Student says: Yes

Decide if the decimal repeats or ends.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$\sqrt{2}$ is about 1.41421... and never repeats or ends. It is irrational.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 0.25 a rational number? Student says: No

Write the decimal as a fraction.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
0.25 = 1/4, so it is rational. Every terminating decimal is rational.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is π a rational number? Student says: Yes

Decide if the decimal repeats or ends.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
π goes on forever without repeating, so it is irrational.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.777… (the 7 repeats forever) as a fraction. Student's answer: 7/10

Let x = 0.777…, then multiply by 10 and subtract.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
7/10 is 0.7, not 0.777…. Since 10x - x = 7, x = 7/9.	**7/9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

the decimal 0.1010010001... (the number of zeros grows, with no repeating pattern) rational? Student says: Y

Look for a repeating pattern.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The decimal never repeats a block of digits, so it cannot be written as a fraction. It is irrational.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Between which two whole numbers is $\sqrt{50}$? Student's answer: between 5 and 6

Find the perfect squares near 50.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$7^2 = 49$ and $8^2 = 64$. Since 50 is just above 49, $\sqrt{50}$ is between 7 and 8.	**Between 7 and 8**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Locate $\sqrt{10}$ on a number line. Student says: It is closer to 4.

Find the nearby whole numbers and compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$\sqrt{10}$ is about 3.16. It is between 3 and 4 and is closer to 3.	**Closer to 3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare $\sqrt{7}$ and 2.6. Student says: $\sqrt{7}$ is less than 2.6.

Square each value to compare.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2.6^2 = 6.76$, which is less than 7. So $\sqrt{7}$ is greater than 2.6.	**$\sqrt{7}$ is greater**

Original task. Format inspired by error analysis routines.

10

Find the Error

8.NS.2

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Compare 3.14 and √9.8. Student says: √9.8 is greater.

Estimate the square root.

What did the student do wrong? Why do you think the student made this error?

√9.8 is about 3.13. Since 3.14 is greater than 3.13, the decimal 3.14 is greater.

Solve the problem correctly. Show all of your work.

3.14 is greater

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Approximate $\sqrt{48}$ to the nearest tenth. Student's answer: 6.8

Estimate and check.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$6.9^2 = 47.61$ and $7.0^2 = 49$. $\sqrt{48}$ is about 6.93, which rounds to 6.9.	**6.9**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Approximate π to the nearest hundredth. Student's answer: 3.1

Look at the thousandths digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
π is about 3.14159. Rounding to the nearest hundredth gives 3.14.	**3.14**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $2^3 \times 2^2$. Student's answer: 2

Use the rule for multiplying powers with the same base.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied the exponents. For the same base, add the exponents: $2^3 \times 2^2 = 2 = 32$.	**2 or 32**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 5. Student's answer: 0

Use the rule for zero exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Any nonzero number to the power 0 equals 1. So 5 = 1.	**1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 10^2. Student's answer: -100

Use the rule for negative exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A negative exponent means the reciprocal. $10^2 = 1/10^2 = 1/100 = 0.01$.	**0.01**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $(3^2)^3$. Student's answer: 3

Use the rule for a power of a power.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the exponents. For a power of a power, multiply: $(3^2)^3 = 3 =$ 729.	**3 or 729**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Evaluate 4^{1}. Student's answer: -4

Use the rule for negative exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A negative exponent gives the reciprocal, not a negative number. $4^{1} = 1/4$.	**1/4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Simplify $x \div x^2$. Student's answer: x

Use the rule for dividing powers with the same base.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the exponents. For division, subtract: $x \div x^2 = x^2$.	**x^2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find $\sqrt{49}$. Student's answer: 24.5

Find the number that times itself is 49.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided 49 by 2. The square root of 49 is 7, since 7 × 7 = 49.	**7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 = 36$ for the positive value of x. Student's answer: $x = 3$

Take the square root of both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student took the square root of 9 instead of 36. Since $6^2 = 36$, $x = 6$.	**6**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find 27. Student's answer: 9

Find the number that times itself three times is 27.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student squared 3 instead of cubing it. 3 × 3 × 3 = 27, so the cube root is 3.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x^2 = 81$. Student's answer: $x = 9$

Find all solutions.

What did the student do wrong? Why do you think the student made this error?

Both 9 and -9 square to 81. The solutions are x = 9 and x = -9.

Solve the problem correctly. Show all of your work.

x = 9 or x = -9

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find $\sqrt{0.25}$. Student's answer: 0.125

Find the number that times itself is 0.25.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student halved 0.25. Since 0.5 × 0.5 = 0.25, the square root is 0.5.	**0.5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find $\sqrt{(-8)}$. Student's answer: no real solution

Think about the sign of a cube.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Cube roots can be negative. $(-2) \times (-2) \times (-2) = -8$, so $\sqrt{(-8)} = -2$.	**-2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 4.5×10^3 in standard form. Student's answer: 450

Move the decimal point.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Multiplying by 10^3 moves the decimal three places right. $4.5 \times 1{,}000 = 4{,}500$.	**4,500**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.00052 in scientific notation. Student writes: 5.2 × 10

Move the decimal point to get a number between 1 and 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The exponent should be negative because the number is small. The answer is 5.2 × 10.	**5.2 × 10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which is greater: 3×10 or 5×10^3? Student's answer: 5×10^3

Compare the powers of 10 first.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$3 \times 10 = 30{,}000$, and $5 \times 10^3 = 5{,}000$. So 3×10 is greater.	**3×10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(2 \times 10^3)(3 \times 10^2)$. Student's answer: 5×10

Multiply the coefficients and add the exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2 and 3 instead of multiplying. $2 \times 3 = 6$, so the answer is 6×10.	**6×10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 6,200,000 in scientific notation. Student writes: 6.2 × 10

Count the places the decimal moves.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The decimal moves 6 places to the left. The answer is 6.2 × 10.	**6.2 × 10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The distance to a star is 9.46×10^{12} km. Is this number written in standard form? Student says: Yes

Check the form.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The number is written in scientific notation, not standard form. Standard form would be 9,460,000,000,000 km.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A city has 2.4 × 10 people. How many people are in 3 cities of the same size? Student's answer: 7.2 × 10

Multiply the coefficient, keep the power of 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student changed the exponent. 3 × 2.4 = 7.2, so the answer is 7.2 × 10.	**7.2 × 10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Divide $(8 \times 10) \div (2 \times 10^2)$. Student's answer: 4×10^3

Divide the coefficients and subtract the exponents.

What did the student do wrong? Why do you think the student made this error?

The student subtracted 2 from 6 incorrectly. 6 − 2 = 4, so the answer is 4 × 10.

Solve the problem correctly. Show all of your work.

4×10

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Multiply $(6 \times 10^3)(5 \times 10^2)$. Student's answer: 3×10

Multiply the coefficients and add the exponents.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
6 × 5 = 30, and −3 + (−2) = −5. So 30 × 10 = 3 × 10.	**3 × 10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Light travels 3×10 km each second. How far does it travel in 2×10^2 seconds? Student's answer: 6×10 km

Multiply the rate by the time.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to add the exponents. $3 \times 2 = 6$, and $10 \times 10^2 = 10$, so the answer is 6×10 km.	**6×10 km**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Write 0.0045 in scientific notation. Student writes: 4.5×10^3

Move the decimal point to the right of the first nonzero digit.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The number is small, so the exponent is negative. The answer is 4.5×10^3.	**4.5×10^3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is 12 × 10 in scientific notation? Student says: Yes

Check that the coefficient is between 1 and 10.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The coefficient 12 is not between 1 and 10. The correct form is 1.2 × 10.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, 0) and (2, 6). What is its slope? Student's answer: 1/3

Use rise over run.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student divided run by rise. Slope = rise ÷ run = 6 ÷ 2 = 3.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line rises 4 units for every 2 units across. What is its slope? Student's answer: 1/2

Use rise over run.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student flipped the ratio. Slope = 4 ÷ 2 = 2.	**2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (1, 5) and (3, 1). What is its slope? Student's answer: 2

Subtract the y-values and the x-values in the same order.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The line goes down as x increases, so the slope is negative. (1 - 5) ÷ (3 - 1) = -4 ÷ 2 = -2.	**-2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which line is steeper: slope 3 or slope 1/2? Student says: slope 1/2

Compare the sizes of the slopes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A larger slope is steeper. 3 is larger than 1/2, so the line with slope 3 is steeper.	**Slope 3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line with slope 0.5 goes through the origin. What is y when x = 10? Student's answer: 20

Multiply the slope by x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 2 instead of 0.5. $y = 0.5 \times 10 = 5$.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which line is increasing: slope -4 or slope 2? Student's answer: slope -4

Think about the sign of the slope.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A positive slope means the line goes up as x increases. The line with slope 2 is increasing.	**Slope 2**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, 0) and (1, 4). Write its equation. Student writes: y = x + 4

Find the slope, then the y-intercept.

What did the student do wrong? Why do you think the student made this error?

The slope is 4 and the y-intercept is 0. The equation is y = 4x.

Solve the problem correctly. Show all of your work.

y = 4x

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line has a slope of 2 and a y-intercept of 3. Write its equation. Student writes: y = 3x + 2

Use y = mx + b.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student swapped the slope and the intercept. m = 2 and b = 3, so y = 2x + 3.	**y = 2x + 3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (0, -2) and (2, 2). Write its equation. Student writes: y = -2x + 2

Find the slope, then the y-intercept.

What did the student do wrong? Why do you think the student made this error?

The slope is (2 - (-2)) ÷ (2 - 0) = 2, not -2.
The equation is y = 2x - 2.

Solve the problem correctly. Show all of your work.

y = 2x - 2

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the point (3, 11) on the line y = 2x + 5? Student says: No

Substitute 3 for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2(3) + 5 = 11, so the point is on the line.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line goes through (1, 5) and (2, 8). Write its equation. Student writes: $y = 3x + 5$

Find the slope, then use a point to find b.

What did the student do wrong? Why do you think the student made this error?

The slope is 3. Using (1, 5): 5 = 3(1) + b, so b = 2. The equation is y = 3x + 2.

Solve the problem correctly. Show all of your work.

y = 3x + 2

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line has a slope of -1/2 and a y-intercept of 4. Write its equation. Student writes: y = 1/2x + 4

Use y = mx + b.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student dropped the negative sign on the slope. The equation is y = -1/2x + 4.	**y = -1/2x + 4**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $3(x + 2) = 2x + 9$. Student's answer: $x = 15$

Distribute, then collect the x terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$3x + 6 = 2x + 9$ gives $x = 3$. The student added the constants incorrectly.	**$x = 3$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2x + 3 = 2x + 5$. Student's answer: $x = 1$

Subtract 2x from both sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Subtracting 2x leaves 3 = 5, which is never true. The equation has no solution.	**No solution**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $4(x + 1) = 4x + 4$. Student's answer: $x = 0$

Distribute, then compare both sides.

What did the student do wrong? Why do you think the student made this error?

Both sides simplify to 4x + 4, which is always true. Every real number is a solution.

Solve the problem correctly. Show all of your work.

Infinitely many solutions

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $5 - (x - 2) = 2$. Student's answer: $x = -5$

Distribute the negative sign to both terms.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not change the sign of the −2. $5 - x + 2 = 2$ gives $7 - x = 2$, so $x = 5$.	**$x = 5$**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $x/3 + 1 = 4$. Student's answer: $x = 3$

Undo the addition, then multiply by 3.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not multiply by 3. $x/3 = 3$, so $x = 9$.	**$x = 9$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve $2(x - 3) = 10$. Student's answer: $x = 6.5$

Distribute, then solve.

What did the student do wrong? Why do you think the student made this error?

The student did not multiply the 3 by 2. $2x - 6 = 10$ gives $2x = 16$, so $x = 8$.

Solve the problem correctly. Show all of your work.

$x = 8$

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = x + 1$ and $y = 3x - 3$. Student's answer: (1, 2)

Set the two expressions equal.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$x + 1 = 3x - 3$ gives $x = 2$, and then $y = 3$. The solution is (2, 3). The student used $x = 1$.	**(2, 3)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $x + y = 10$ and $x - y = 4$. Student's answer: (3, 7)

Add the equations to eliminate y.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Adding gives 2x = 14, so x = 7 and y = 3. The student swapped the values.	**(7, 3)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Solve the system $y = 2x + 1$ and $y = 2x + 5$. Student's answer: (1, 3)

Compare the two equations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Both lines have slope 2 but different intercepts, so they are parallel. There is no solution.	**No solution**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is (2, 3) a solution of the system $y = x + 1$ and $y = 3x - 3$? Student says: No

Substitute into both equations.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$2 + 1 = 3$ and $3(2) - 3 = 3$. The point works in both equations.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The lines y = 2x and y = -x + 6 cross at a point. What is the point? Student's answer: (4, 2)

Set the two expressions equal and solve for x.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2x = -x + 6 gives x = 2 and y = 4. The student swapped the coordinates.	**(2, 4)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

5 + 2x dollars and Plan B costs 3x dollars. At what number of items do the plans cost the same? Student's

Set the two costs equal.

What did the student do wrong? Why do you think the student made this error?

5 + 2x = 3x gives x = 5. At 5 items, both cost $15.

Solve the problem correctly. Show all of your work.

5 items

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is the relation {(1, 2), (1, 3), (2, 4)} a function? Student says: Yes

Check whether any input has two outputs.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The input 1 is paired with both 2 and 3. A function gives exactly one output for each input.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is y = 2x a function? Student says: No

Check that each input gives one output.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each x gives exactly one y, so y = 2x is a function.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A graph crosses a vertical line in two places. Is the graph a function? Student says: Yes

Use the vertical line test.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Two crossings mean one input has two outputs, so it fails the vertical line test. It is not a function.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table maps input 3 to output 9 and input 4 to output 9. Is this a function? Student says: No

Check each input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each input has only one output. Two inputs can share the same output, so this is a function.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A relation maps input 5 to outputs 6 and 7. Is it a function? Student says: Yes

Check each input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Input 5 has two outputs, so it is not a function.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is y = |x| a function? Student says: No

Check that each input gives one output.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each x gives one absolute value. So y = |x| is a function.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

rate of change of 3 from a table. Function B is y = 2x + 3. Which has the greater rate of change? Student's a

Compare the rates of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Function A has a rate of 3, and Function B has a rate of 2. Function A is greater.	**Function A**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

y = 3x + 1. Function B has the points (0, 2) and (1, 5). Which has the greater starting value? Student's answe

Find the starting value of each.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Function A starts at 1. Function B starts at 2, so Function B has the greater starting value.	**Function B**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s y = 5x. Function B has the points (0, 0) and (1, 4). Which has the greater rate of change? Student's answer

Compare the slopes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Function A has a rate of 5. Function B has a rate of 4. Function A is greater.	**Function A**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = 2x^2$ a linear function? Student says: Yes

Check whether the rate of change is constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The x is squared, so the rate of change is not constant. It is not linear.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has the points (1, 3), (2, 6), and (3, 10). Is it linear? Student says: Yes

Check whether the rate of change is constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The changes are 3 and then 4. The rate is not constant, so the table is not linear.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

tance d = 40t. Car B has the points (1, 30) and (2, 60) for time and distance. Which car is faster? Student's

Compare the rates of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Car B's rate is 30 miles per hour. Car A's rate is 40 miles per hour, so Car A is faster.	**Car A**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find y when $y = x^2$ and $x = 3$. Student's answer: 6

Square the input.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student multiplied by 2. $3^2 = 9$, so y = 9.	**9**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which function is nonlinear: $y = 4x + 1$ or $y = x^2$? Student's answer: $y = 4x + 1$

Look for the variable raised to a power.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$y = 4x + 1$ has a constant rate of change, so it is linear. $y = x^2$ is nonlinear.	**$y = x^2$**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The table has x = 1, 2, 3 and y = 1, 4, 9. Is the relationship linear? Student says: Yes

Check whether the rate of change is constant.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The changes are 3 and then 5. The rate is not constant, so it is not linear.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

The area of a square is $A = s^2$, where s is the side length. Is this relationship linear? Student says: Yes

Check for a squared variable.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The side is squared, so the relationship is not linear.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is $y = 1/x$ a linear function? Student says: Yes

Check the rate of change.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The graph curves and the rate of change is not constant, so it is nonlinear.	**No**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Which way does the graph of $y = x^2$ open? Student's answer: downward

Think about the sign in front of x^2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The coefficient 1 is positive, so the parabola opens upward.	**Upward**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A table has the points (0, 3) and (2, 7). Write the linear function. Student writes: $y = 3x + 2$

Find the rate of change and the initial value.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The rate of change is (7 - 3) ÷ 2 = 2, and the initial value is 3. The function is y = 2x + 3.	**y = 2x + 3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

What is the initial value of y = 5x - 4? Student's answer: 5

Find the value of y when x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
5 is the rate of change. The initial value is -4, because y = -4 when x = 0.	**-4**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

k has 10 gallons and loses 2 gallons each hour. Write the function. Student's answer: starting value 10 and r

Identify the starting amount and the rate.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The rate is the amount lost each hour, which is -2. The function is y = 10 - 2x.	**Rate -2, starting value 10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A savings account starts with $50 and gains $15 each week. What is the initial value? Student's answer: 15

Find the starting amount.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
15 is the weekly gain, which is the rate. The initial value is $50.	**$50**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A line passes through (0, 6) and (3, 0). What is the rate of change? Student's answer: 2

Use (change in y) ÷ (change in x).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The y-values go down by 6 over 3 steps, so the rate is (0 - 6) ÷ (3 - 0) = -2.	**-2**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the rate of change for the points (1, 4) and (3, 10). Student's answer: 6

Use (change in y) ÷ (change in x).

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the change in y alone. **(10 − 4) ÷ (3 − 1) = 6 ÷ 2 = 3.**	**3**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

oh starts high on the left and falls steadily to the right. Is it increasing or decreasing? Student's answer: incre

Look at the direction of the graph from left to right.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The graph goes down as x increases, so it is decreasing.	**Decreasing**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

at for a while and then rises steadily. Which description fits the flat part? Student's answer: increasing at a c

Think about what a flat graph means.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A flat part shows zero rate of change, so the quantity is constant there.	**Constant**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

re graph rises from 40 to 80 and then falls to 50. When is the temperature decreasing? Student's answer: f

Find the part of the graph going down.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The temperature rises from 40 to 80. It falls after reaching 80, when it decreases to 50.	**After it reaches 80**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A graph is a curve that gets steeper as x increases. Student says: The rate of change is constant.

Think about whether the steepness changes.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The graph is steeper at larger x, so the rate of change is not constant.	**No, the rate changes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ws distance rising to 12 miles at 2 hours, then staying at 12 for 1 hour. What happens in the last hour? Studen

Look at the shape of the graph in that hour.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The distance stays at 12, so the rider is not moving forward. The rider stopped.	**The rider stopped**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A graph starts at the point (0, 5). What is the initial value? Student's answer: 0

Find y when x = 0.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The graph starts at y = 5 when x = 0, so the initial value is 5.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Translate the point (2, 3) right 4 units. Student's answer: (2, 7)

Move the x-value for a horizontal shift.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student moved up instead of right. Right 4 gives (6, 3).	**(6, 3)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Reflect the point (2, 3) across the x-axis. Student's answer: (-2, 3)

Think about what changes in a reflection across the x-axis.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Reflecting across the x-axis changes the sign of y, not x. The answer is (2, -3).	**(2, -3)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Reflect the point (2, 3) across the y-axis. Student's answer: (2, -3)

Think about what changes in a reflection across the y-axis.

What did the student do wrong? Why do you think the student made this error?

Reflecting across the y-axis changes the sign of x. The answer is (-2, 3).

Solve the problem correctly. Show all of your work.

(-2, 3)

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rotate the point (1, 0) 90° counterclockwise about the origin. Student's answer: (1, 0)

Turn the point a quarter turn counterclockwise.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not rotate the point. A 90° counterclockwise turn moves (1, 0) to (0, 1).	**(0, 1)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (2, 3) by a scale factor of 2 from the origin. Student's answer: (2, 6)

Multiply both coordinates by the scale factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student only changed the y-value. Both coordinates are multiplied: (4, 6).	**(4, 6)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Translate the point (5, 5) left 3 units and down 2 units. Student's answer: (8, 7)

Subtract from x and y.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Left and down mean subtract. 5 - 3 = 2 and 5 - 2 = 3, so the answer is (2, 3).	**(2, 3)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are two rectangles that are both 3 by 5 congruent, with one rotated? Student says: No

Check whether the size and shape match.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Rotation keeps size and shape. The rectangles are congruent.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are triangles with sides 3, 4, 5 and 4, 5, 6 congruent? Student says: Yes

Compare the side lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The side lengths do not match, so the triangles are not congruent.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle is reflected across a line to make a new triangle. Are they congruent? Student says: No

Think about what a reflection does.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A reflection keeps size and shape, so the triangles are congruent.	**Yes**

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Find the Error

8.G.2

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

nsformation maps a figure to a congruent figure: translation, reflection, rotation, or dilation? Student's answ

Think about which ones keep size.

What did the student do wrong? Why do you think the student made this error?

A dilation changes the size. Translation, reflection, and rotation keep the figure congruent.

Solve the problem correctly. Show all of your work.

Translation, reflection, or rotation

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a square with side 4 congruent to a square with side 8? Student says: Yes

Compare the side lengths.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The sides are different lengths, so the squares are not congruent.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Two triangles have the same three angles but different sizes. Are they congruent? Student says: Yes

Think about size as well as angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Matching angles only means the triangles are similar. Congruent figures must also have the same size.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (3, -1) by a scale factor of 2 from the origin. Student's answer: (5, 1)

Multiply both coordinates by 2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 2 to each coordinate. Multiply: (6, -2).	**(6, -2)**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (6, 10) by a scale factor of 1/2 from the origin. Student's answer: (3, 4)

Multiply both coordinates by 1/2.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student changed the y-value wrong. 10 × 1/2 = 5, so the answer is (3, 5).	**(3, 5)**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

pe has a side of 2 units. It is dilated by a scale factor of 3. What is the new side length? Student's answer: 5

Multiply the side by the scale factor.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 3 to 2. Multiply: 2 × 3 = 6 units.	**6 units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Dilate the point (2, 3) by a scale factor of 2 from the center (1, 1). Student's answer: (4, 6)

Find the distance from the center and double it.

What did the student do wrong? Why do you think the student made this error?

The student doubled the coordinates instead of the distance from (1, 1). The distance is 1 in x and 2 in y, which doubles to 2 and 4. The answer is (3, 5).

Solve the problem correctly. Show all of your work.

(3, 5)

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A figure is dilated by a scale factor of 0.5. Student says: The new figure is larger.

Compare the scale factor to 1.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
A scale factor less than 1 makes the figure smaller.	**No, it is smaller**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A figure is dilated by a scale factor of 3. Student says: The area becomes 3 times as large.

Think about how area scales.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Area scales by the square of the scale factor. The area becomes $3^2 = 9$ times as large.	**No, 9 times as large**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Triangles have sides 3, 4, 5 and 6, 8, 10. Are they similar? Student says: No

Check whether the sides have the same ratio.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Each side in the second triangle is 2 times the first. The triangles are similar.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

es a side of 6 in a similar triangle. The matching side is 10 in the first triangle. What is the matching side in t

Set up a proportion.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The ratio is 4/6 = 2/3. So 10/x = 2/3, and x = 15.	**15**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rectangles are 2 by 3 and 4 by 6. Are they similar? Student says: No

Compare the ratios of the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
2/4 = 3/6 = 1/2. The rectangles are similar.	**Yes**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Are two triangles similar if two pairs of angles match? Student says: No, you need the sides too.

Think about angle-angle similarity.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Two matching pairs of angles are enough for similarity. The third angles must match too, so AA is enough.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Rectangles are 3 by 5 and 6 by 9. Are they similar? Student says: Yes

Compare the ratios of the sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
3/6 = 1/2, but 5/9 is not 1/2. The rectangles are not similar.	**No**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

with sides 3 and 9 are similar. What is the scale factor from the small square to the large one? Student's ar

Divide the large side by the small side.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student flipped the ratio. 9 ÷ 3 = 3, so the scale factor is 3.	**3**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles 50° and 60°. What is the third angle? Student's answer: 80°

Angles in a triangle add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted incorrectly. 180 - 110 = 70°.	**70°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

parallel lines are cut by a transversal. One alternate interior angle is 65°. What is the other? Student's answe

Think about alternate interior angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Alternate interior angles are equal. The other angle is 65°.	**65°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ıre cut by a transversal. A corresponding angle is 40°. What is the matching corresponding angle? Student'

Think about corresponding angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Corresponding angles are equal. The matching angle is 40°.	**40°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

n exterior angle of 120° and one remote interior angle of 70°. What is the other remote interior angle? Stude

An exterior angle equals the sum of the two remote angles.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 120 and 70. The other remote angle is 120 - 70 = 50°.	**50°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A triangle has angles 40°, x, and 100°. What is x? Student's answer: 140°

Angles in a triangle add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
40 + x + 100 = 180 gives x = 40°. The student did not subtract the known angles from 180.	**40°**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

s are cut by a transversal. Same-side interior angles include one angle of 110°. What is the other? Student's

Same-side interior angles add to 180°.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
Same-side interior angles are supplementary. 180 - 110 = 70°.	**70°**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has legs 3 and 4. What is the hypotenuse? Student's answer: 7

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the legs. $3^2 + 4^2 = 25$, so $c = 5$.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has legs 6 and 8. What is the hypotenuse? Student's answer: 14

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the legs. $6^2 + 8^2 = 100$, so $c = 10$.	**10**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has a hypotenuse of 13 and a leg of 5. What is the other leg? Student's answer: 8

Use $a^2 + b^2 = c^2$, solving for the missing leg.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 5 from 13. $13^2 - 5^2 = 169 - 25 = 144$, so the leg is 12.	**12**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has a hypotenuse of 15 and a leg of 9. What is the other leg? Student's answer: 6

Use $a^2 + b^2 = c^2$, solving for the missing leg.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 9 from 15. $15^2 - 9^2 = 225 - 81 = 144$, so the leg is 12.	**12**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a triangle with sides 5, 12, and 13 a right triangle? Student says: No

Check whether $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$5^2 + 12^2 = 25 + 144 = 169 = 13^2$. It is a right triangle.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A square has a side of 4. What is the length of its diagonal? Student's answer: 8

Use the Pythagorean theorem with the two sides.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The diagonal is the hypotenuse of a right triangle with legs 4 and 4. √(16 + 16) = √32 = 4√2, which is about 5.66.	**4√2, about 5.66**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A rectangular floor is 12 feet by 5 feet. What is the length of the diagonal? Student's answer: 17 feet

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the sides. $12^2 + 5^2 = 169$, so the diagonal is 13 feet.	**13 feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

ler 10 feet long leans on a wall with its base 6 feet from the wall. How high does it reach? Student's answer:

Use $a^2 + b^2 = c^2$, solving for the height.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 10 and 6. $10^2 - 6^2 = 64$, so the height is 8 feet.	**8 feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A box is 3 by 4 by 12 units. What is the length of its space diagonal? Student's answer: 19

Use the 3D version of the Pythagorean theorem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the sides. The space diagonal is $\sqrt{(3^2 + 4^2 + 12^2)} = \sqrt{169} = 13$ units.	**13 units**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has a hypotenuse of 17 and a leg of 8. What is the other leg? Student's answer: 9

Solve for the missing leg.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student subtracted 8 from 17. $17^2 - 8^2 = 289 - 64 = 225$, so the leg is 15.	**15**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

eet from a point on the ground. A wire runs from the top of the pole to that point. How long is the wire? Stud

Use $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 9 and 12. $9^2 + 12^2 = 225$, so the wire is 15 feet.	**15 feet**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Is a triangle with sides 6, 8, and 10 a right triangle? Student says: No

Check whether $a^2 + b^2 = c^2$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
$6^2 + 8^2 = 36 + 64 = 100 = 10^2$. It is a right triangle.	**Yes**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (0, 0) and (6, 8). Student's answer: 14

Use the distance formula or the Pythagorean theorem.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 6 and 8. The distance is $\sqrt{(6^2 + 8^2)} = \sqrt{100} = 10$.	**10**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (1, 1) and (4, 5). Student's answer: 7

Find the horizontal and vertical distances.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 3 and 4. The distance is $\sqrt{(3^2 + 4^2)} = 5$.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (-2, 3) and (1, -1). Student's answer: 7

Find the horizontal and vertical distances.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The horizontal distance is 3 and the vertical distance is 4. The distance is √(9 + 16) = 5.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (2, 2) and (2, 7). Student's answer: 7

Find the vertical distance.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The points are on a vertical line. The distance is 7 − 2 = 5.	**5**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A right triangle has vertices (0, 0), (3, 0), and (0, 4). What is the length of its hypotenuse? Student's answer: 7

Find the distance between the two vertices that are not on the same axis.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added the legs. The hypotenuse goes from (3, 0) to (0, 4), so its length is $\sqrt{(9 + 16)} = 5$.	**5**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the distance between (0, 0) and (5, 12). Student's answer: 17

Use the distance formula.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student added 5 and 12. The distance is $\sqrt{(25 + 144)} = \sqrt{169} = 13$.	**13**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cylinder with radius 2 and height 3. Student's answer: 6π

Use $V = \pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot to square the radius. $V = \pi \times 2^2 \times 3 = 12\pi$, which is about 37.7 cubic units.	**12π, about 37.7**

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a cone with radius 3 and height 4. Student's answer: 36π

Use $V = (1/3)\pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student forgot the 1/3. $V = (1/3) \times \pi \times 9 \times 4 = 12\pi$, which is about 37.7 cubic units.	**12π, about 37.7**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

Find the volume of a sphere with radius 3. Student's answer: 12π

Use $V = (4/3)\pi r^3$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student did not cube the radius. $V = (4/3) \times \pi \times 27 = 36\pi$, which is about 113.1 cubic units.	**36π, about 113.1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cone has radius 2 and height 6. What is its volume? Student's answer: 24π

Use $V = (1/3)\pi r^2 h$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the cylinder volume. The cone is one third of that, so $V = (1/3) \times \pi \times 4 \times 6 = 8\pi$, about 25.13 cubic units.	**8π, about 25.13**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A sphere has a diameter of 6. What is its volume? Student's answer: 288π

Find the radius first, then use $V = (4/3)\pi r^3$.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student used the diameter as the radius. The radius is 3, so $V = (4/3)\pi \times 27 = 36\pi$.	**36π, about 113.1**

Original task. Format inspired by error analysis routines.

Find the Error

Name: Answer Key Date: _____

Directions: Analyze the student's work shown below. Identify the error. Then correctly solve the problem.

A cylinder has a volume of 50π and a radius of 5. What is its height? Student's answer: 10

Solve $V = \pi r^2 h$ for h.

What did the student do wrong? Why do you think the student made this error?	Solve the problem correctly. Show all of your work.
The student found the wrong height. $50\pi = \pi \times 25 \times h$, so $h = 2$.	**2**

Original task. Format inspired by error analysis routines.